MYSQL ASSIGNMENT 1 – DDL COMMANDS & CONSTRAINTS

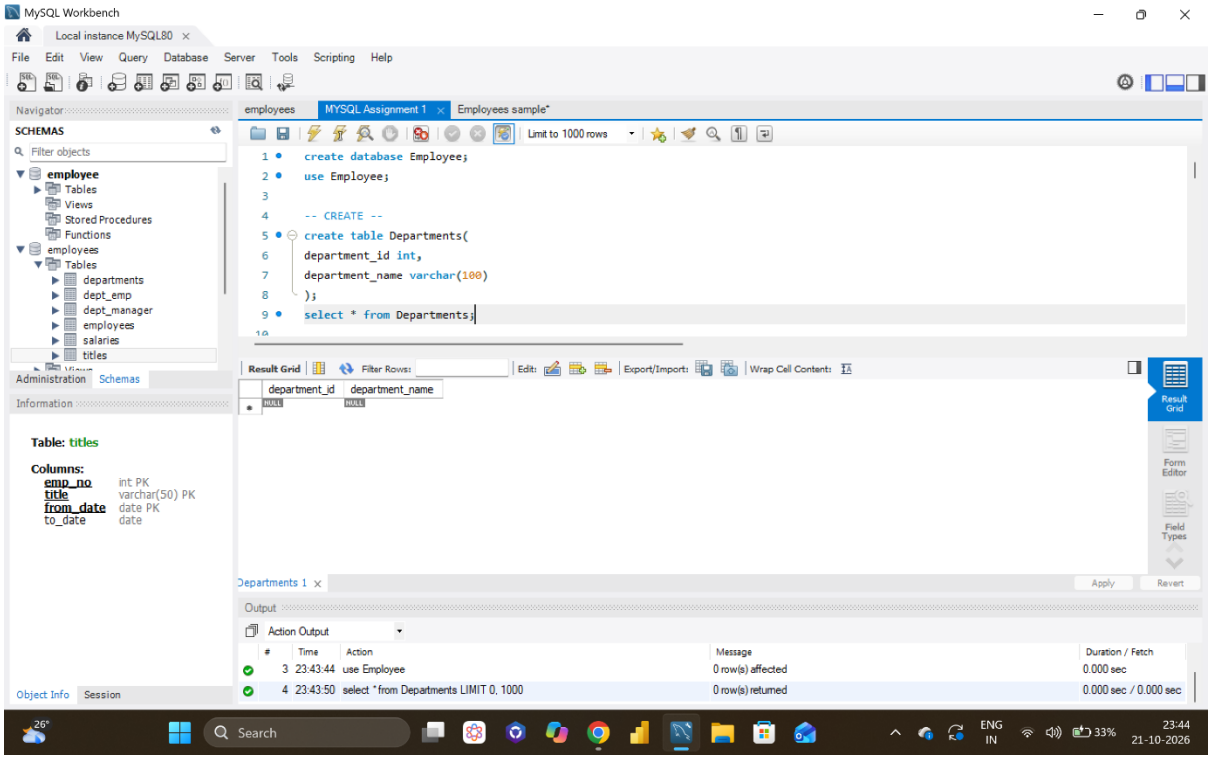
DDL Commands:

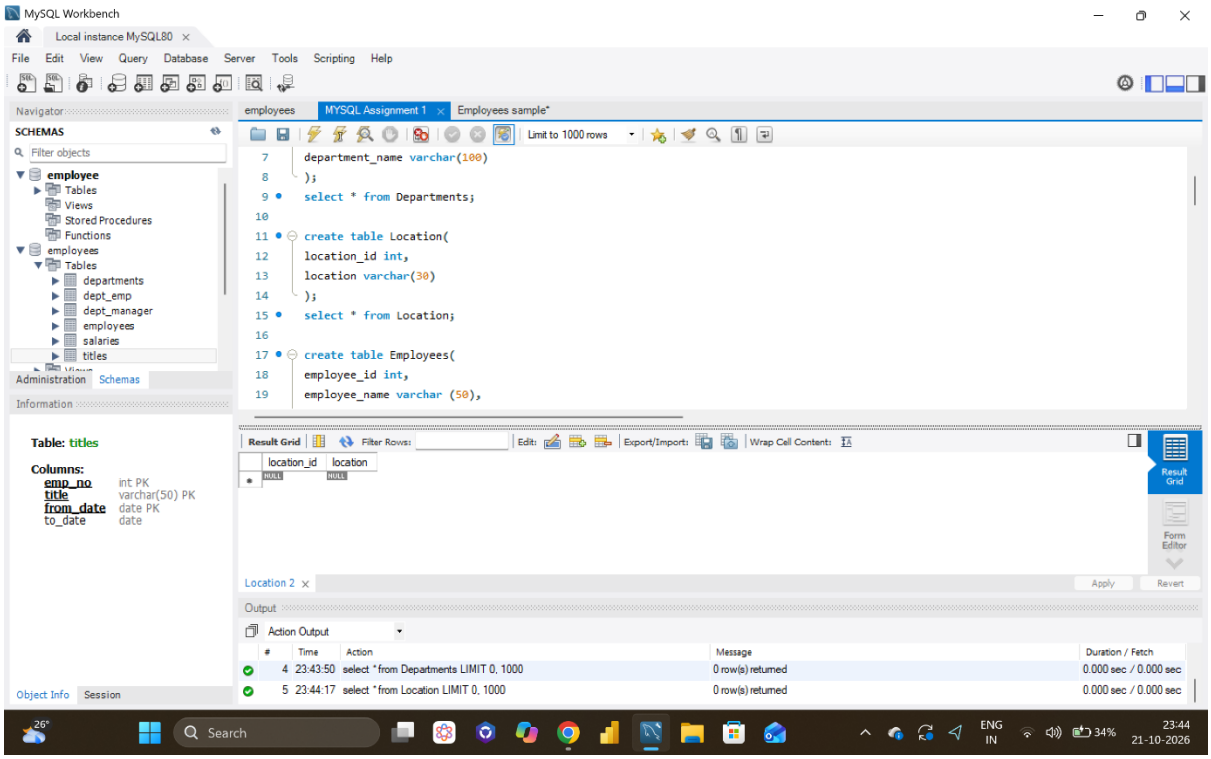
1. Table Creation (CREATE): Write the SQL statements to create a database named "employee" and the following tables based on the provided schema:

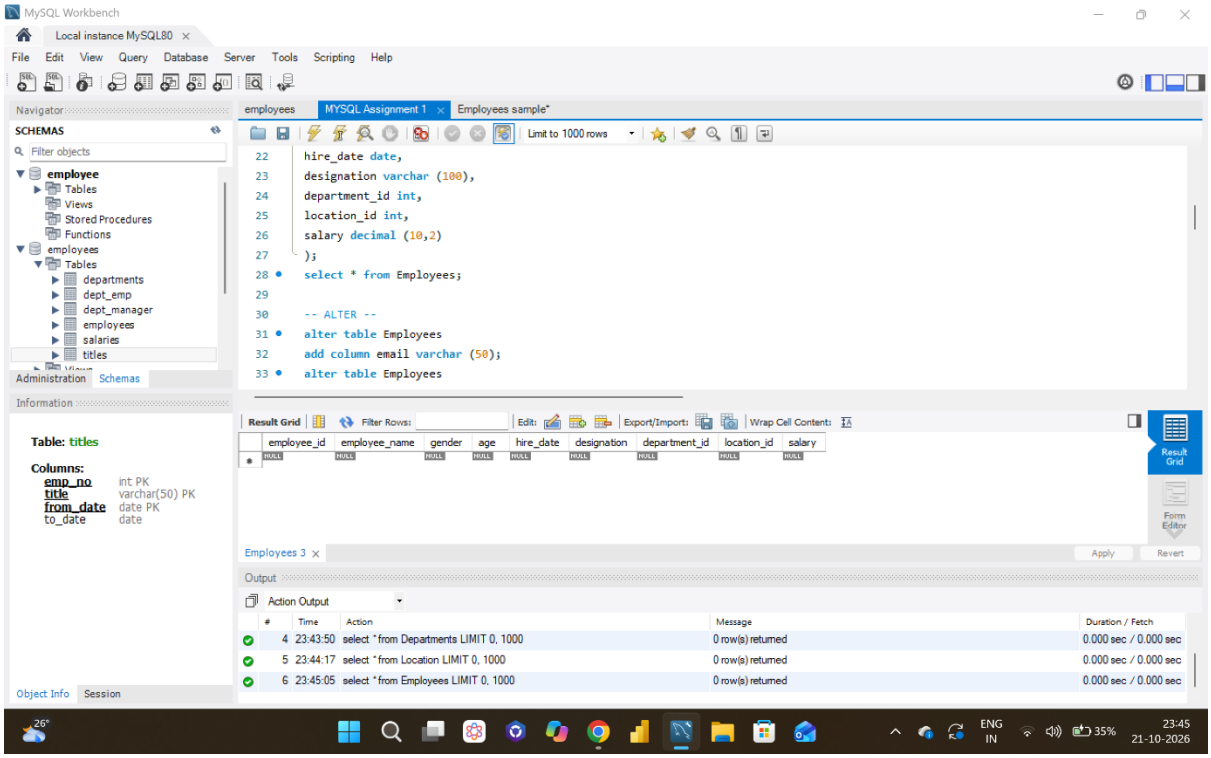
◉ Departments

◉ Location

◉ Employees







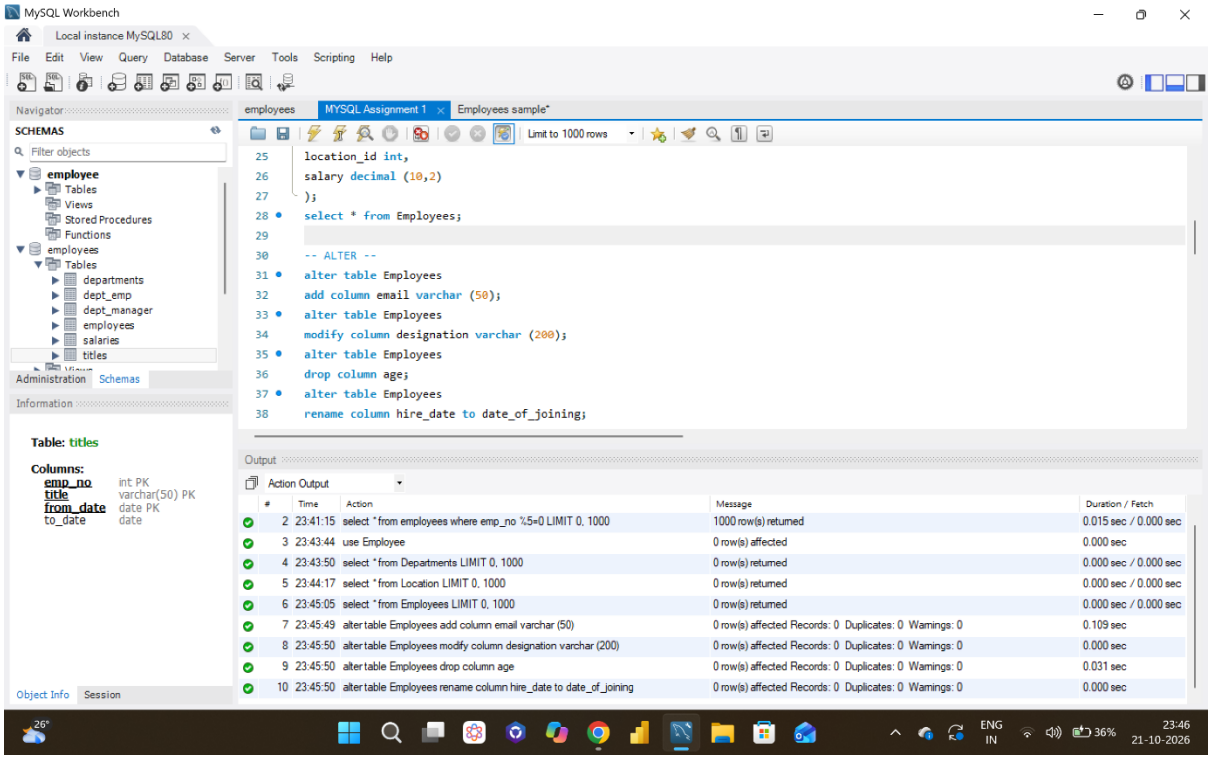
2. Table Alteration (ALTER): Consider the following scenarios and write the SQL statements to alter the structure of the tables accordingly:

◉ Add a new column named "email" to the Employees table to store employee email addresses.

◉ Modify the data type of the "designation" column in the Employees table to support a wider range of values.

◉ Drop the “age” column from the Employees table.

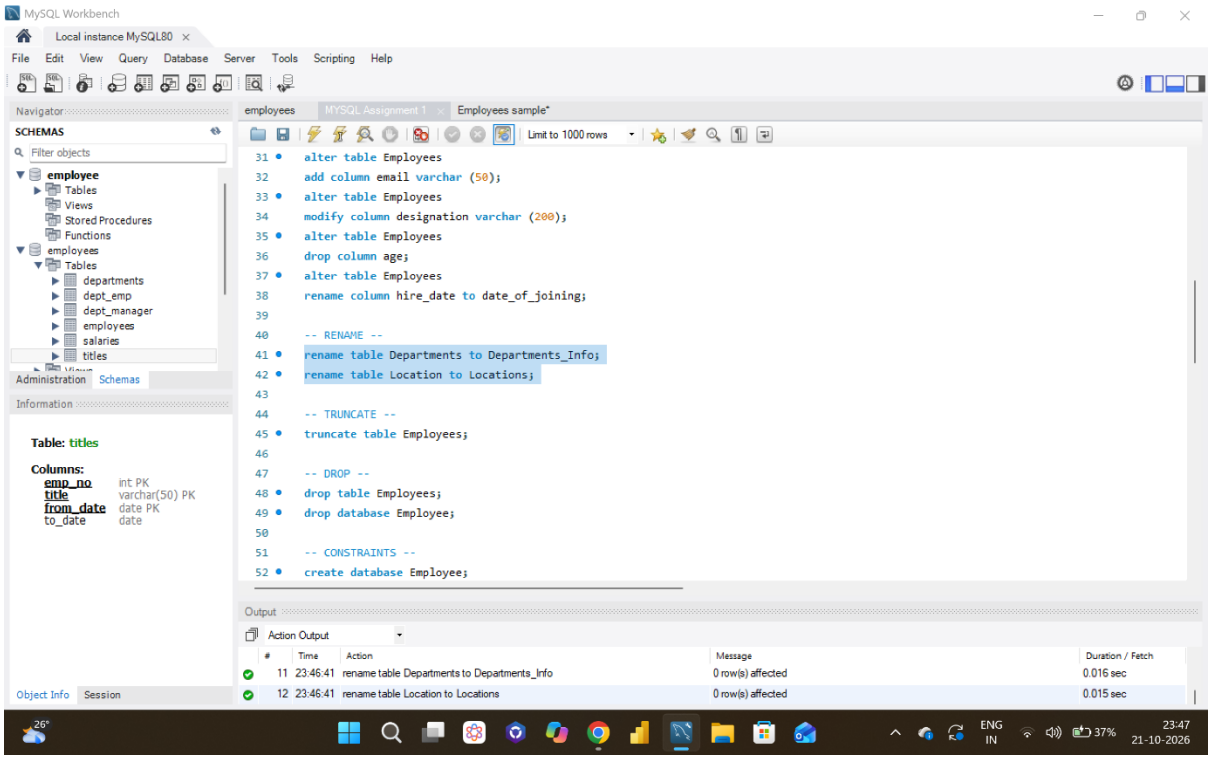
◉ Rename the “hire_date” column to “date_of_joining”.



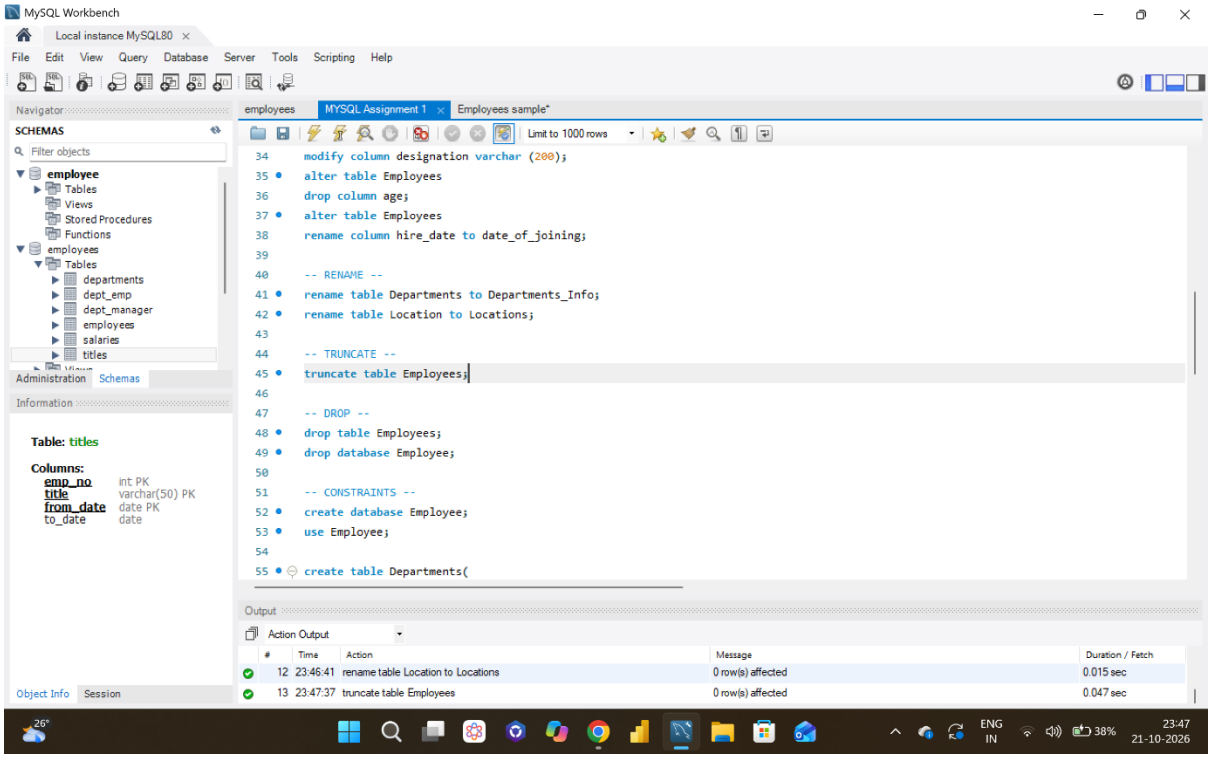
3. Table Renaming (RENAME): Rewrite the SQL statements to rename the following tables:

◉ Rename the "Departments" table to "Departments_Info".

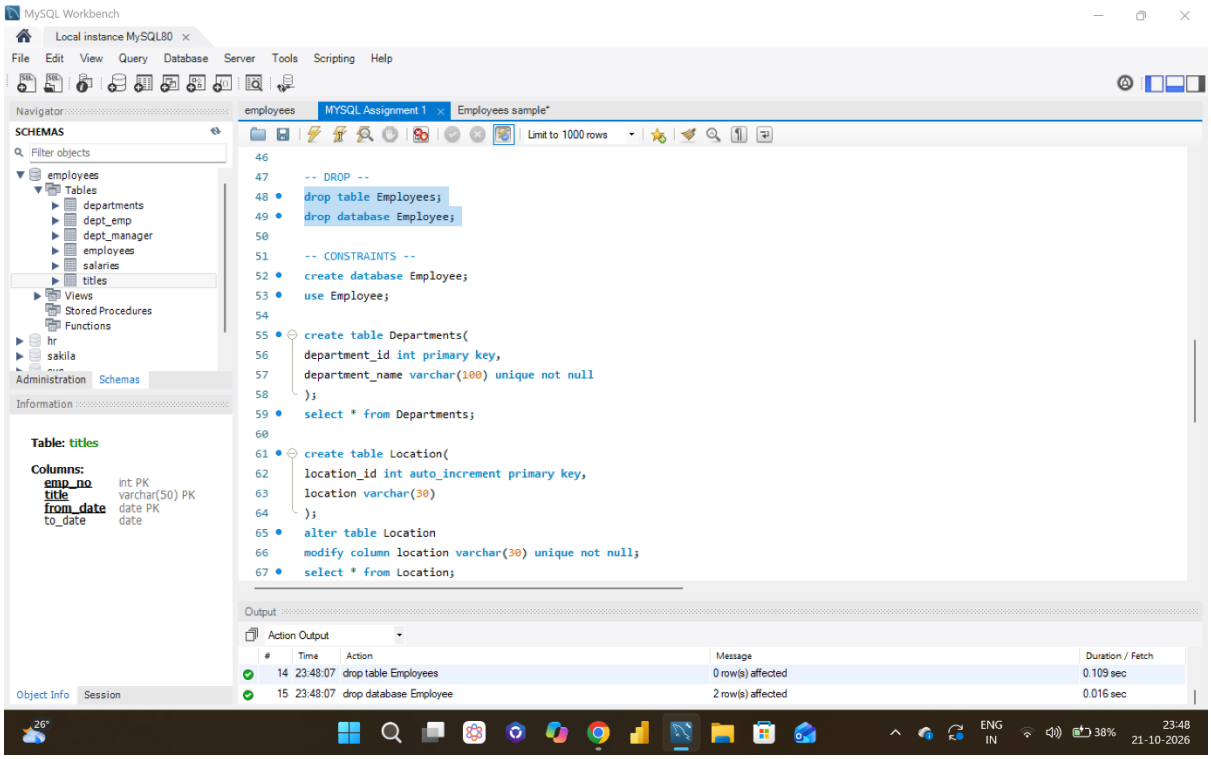
◉ Rename the "Location" table to "Locations".



4. Table Truncation (TRUNCATE): Write an SQL statement to truncate the Employees table.



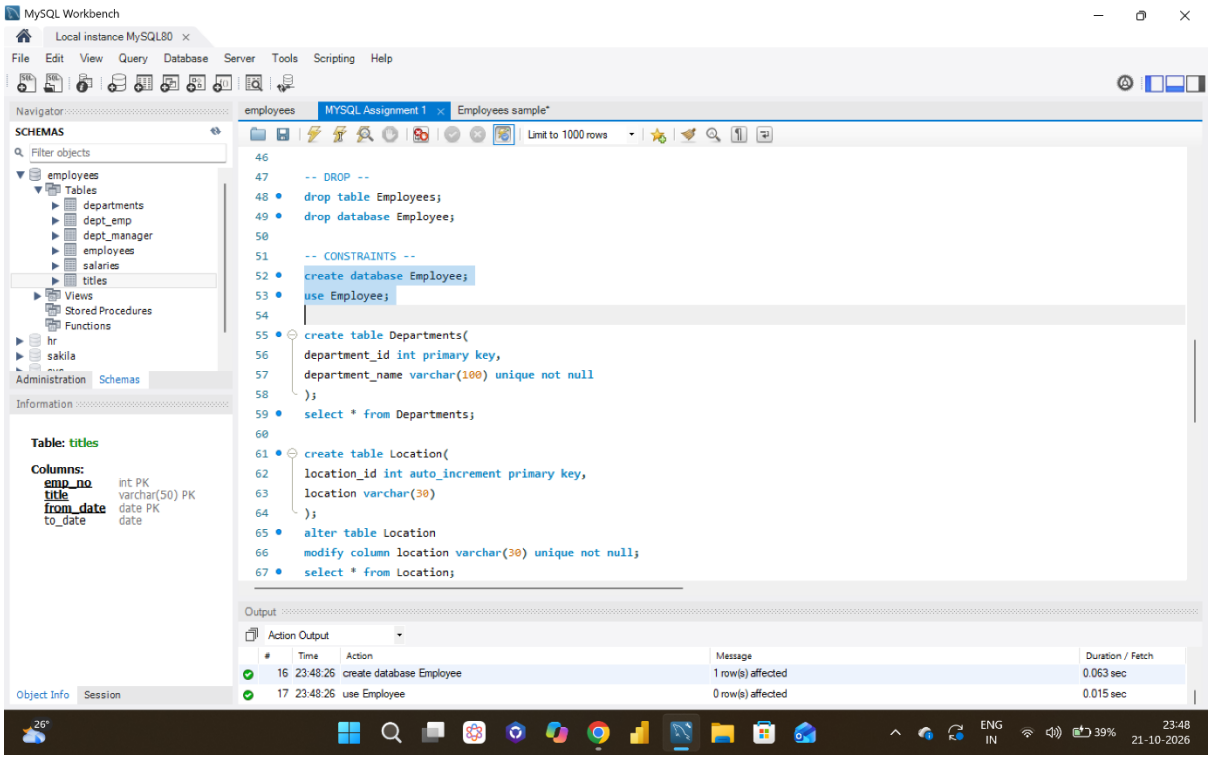
5. Database & Table Dropping (DROP): Write the SQL statements to drop the Employees table and then the “employee” database.



Constraints:

1. Database Recreation:

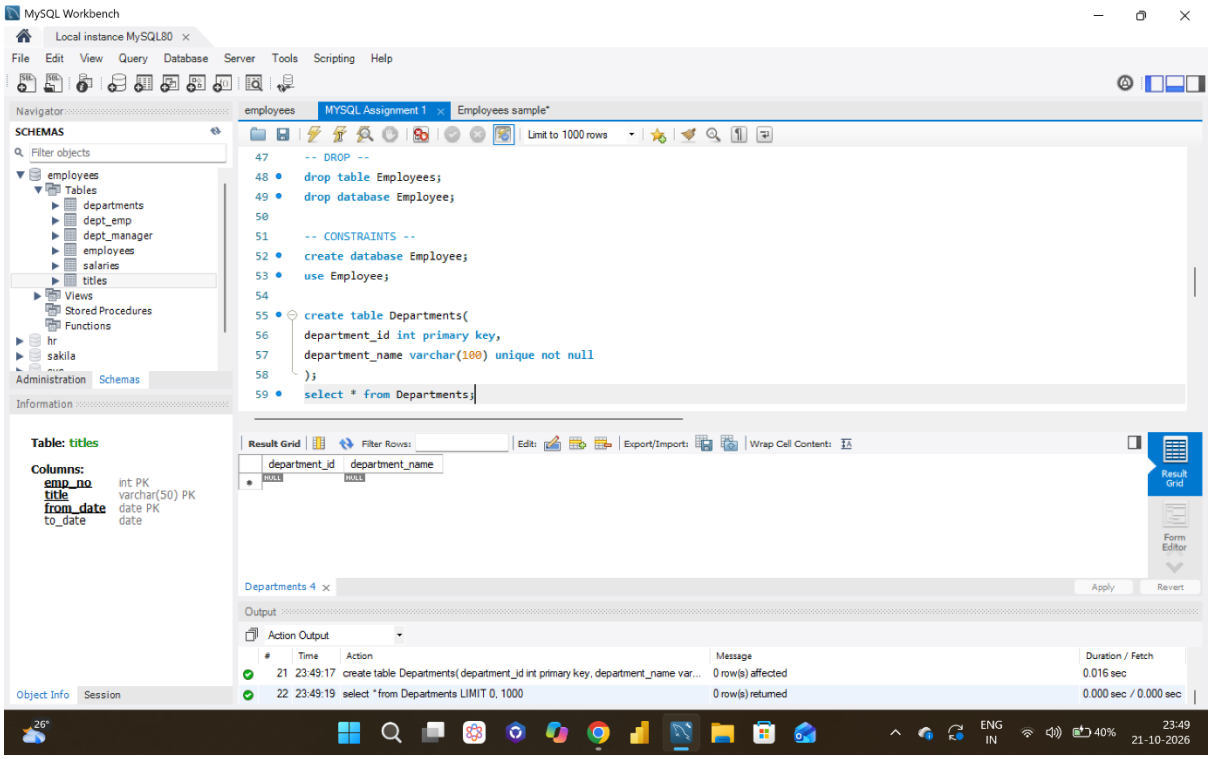
◉ Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.



2. Departments Table:

◉ Ensure that the "department_id" uniquely identifies each department.

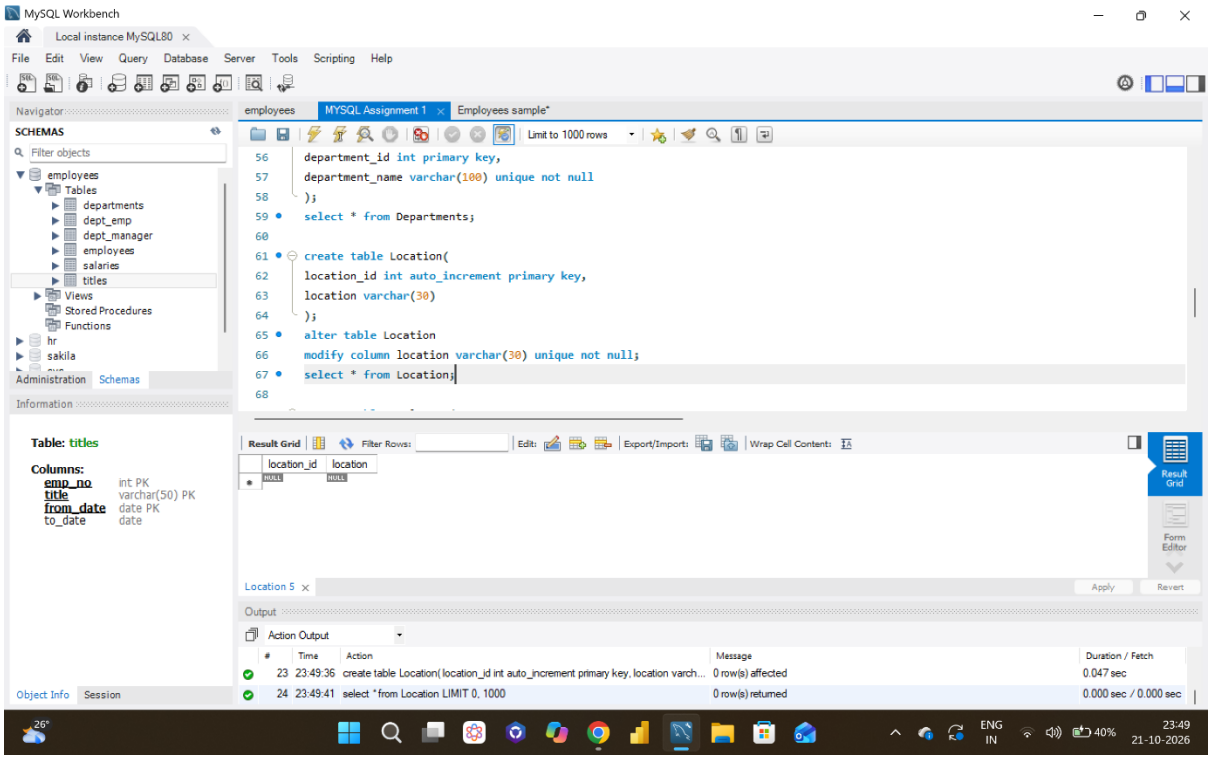
◉ Set up constraints on the "department_name" to avoid duplicate and null entries.



3. Location Table:

◉ Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.

◉ Implement constraints to prevent the insertion of null and duplicate locations.



4. Employees Table:

◉ Guarantee that each employee has a distinct identifier.

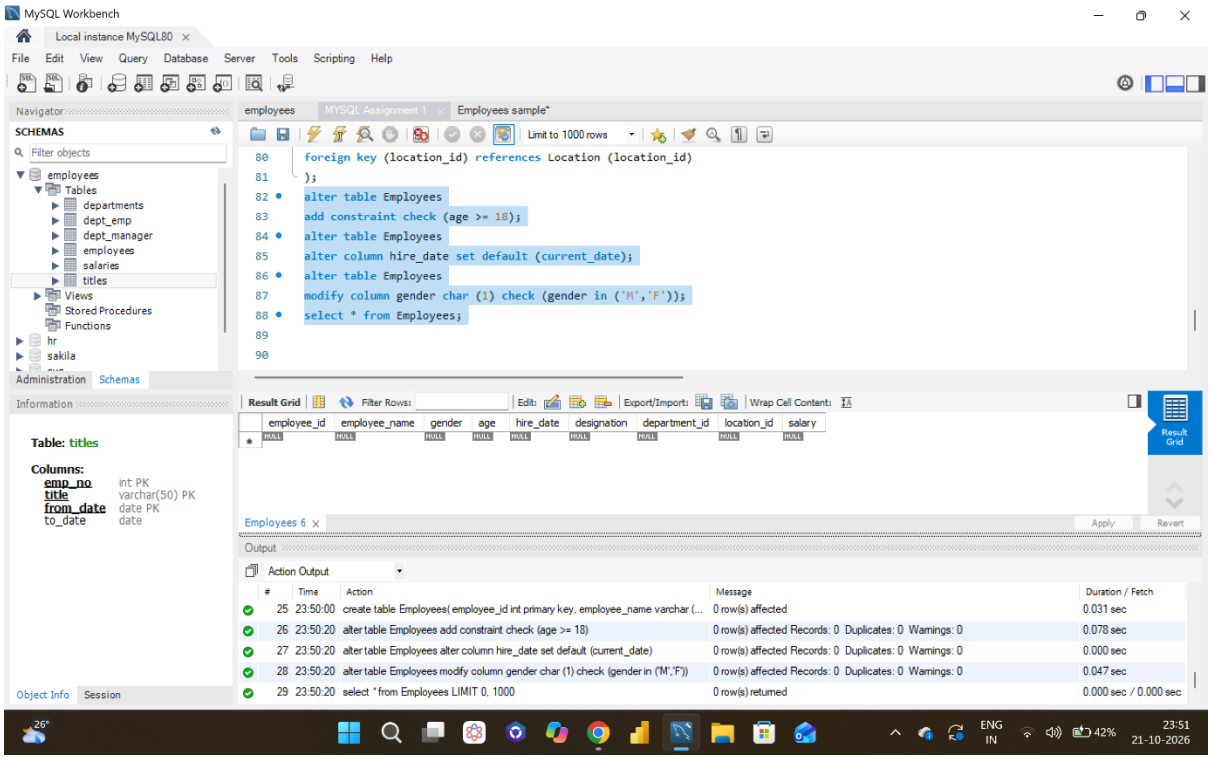
◉ Create a restriction to ensure that the employee's name is always provided.

◉ Limit the acceptable values for the "gender" field to only 'M' or 'F'.

◉ Enforce a condition to ensure that the employee's age is 18 or above.

◉ Automatically assign the current date to the "hire_date" field if not specified.

◉ Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employees
Tables
departments
dept_emp
dept_manager
employees
salaries
titles
Views
Stored Procedures
Functions
hr
sakila
Administration Schemas
Information
Table: titles
Columns:
emp_no int PK
title varchar(50) PK
from_date date PK
to_date date
Object Info Session
employees MYSQL Assignment 1 Employees sample*
Limit to 1000 rows
80 foreign key (location_id) references Location (location_id)
81);
82 alter table Employees
83 add constraint check (age >= 18);
84 alter table Employees
85 alter column hire_date set default (current_date);
86 alter table Employees
87 modify column gender char (1) check (gender in ('M','F'));
88 select * from Employees;
89
90
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
Result Grid
Employees 6
Apply Revert
Output
Action Output
Time Action Message Duration / Fetch
25 23:50:00 create table Employees(employee_id int primary key, employee_name varchar (... 0 row(s) affected 0.031 sec
26 23:50:20 alter table Employees add constraint check (age >= 18) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.078 sec
27 23:50:20 alter table Employees alter column hire_date set default (current_date) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec
28 23:50:20 alter table Employees modify column gender char (1) check (gender in ('M','F')) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.047 sec
29 23:50:20 select * from Employees LIMIT 0, 1000 0 row(s) returned 0.000 sec / 0.000 sec
26°
ENG IN
42%
23:51
21-10-2026